Signals and Systems Lecture-15: Laplace Transform 2

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Inverse Laplace Transform (1)

- A common problem in signals and systems is to find the inverse transform of a s-domain function.
- The most common type of Laplace-transform expressions is a ratio of polynomials in s:

$$G(s) = \frac{b_0 + b_1 s + \cdots + b_m s^m}{a_0 + a_1 s + \cdots + a_n s^n}$$

- The numerator and the denominator of $G(s)$ can be factorized resulting in:

$$G(s) = \frac{b_m}{a_n} \frac{(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)}$$

where:

  - $z_1, \ldots, z_m$ are called the zeros of $G(s)$.
  - $p_1, \ldots, p_n$ are called the poles of $G(s)$.
  - In what follows, we assume that:

$$z_i \neq p_j \;\; ; \;\; i = 1, \ldots, m \; ; \; j = 1, \ldots, n$$

Inverse Laplace Transform (2)

Case 1: Assume (for now) that:

- $m < n$.
- No poles are repeated (all roots of the denominator of $G(s)$ have multiplicity one).

In this case, we can write $G(s)$ as:

$$G(s) = \frac{r_1}{s - p_1} + \frac{r_2}{s - p_2} + \cdots + \frac{r_n}{s - p_n}$$

called the Partial Fraction Expansion of $G(s)$.
Given that $e^{-\alpha t}u(t) \leftrightarrow \frac{1}{s+\alpha}$:

$$\begin{aligned} g(t) &= \mathcal{L}^{-1}\left[G(s)\right] \\ &= \left[r_1 e^{p_1 t} + r_2 e^{p_2 t} + \cdots + r_n e^{p_n t}\right] u(t) \end{aligned}$$

Inverse Laplace Transform (3)

$r_1, \ldots, r_n$ can be found by a convenient method:

$$\frac{b_0 + b_1 s + \cdots + b_m s^m}{(s-p_1)(s-p_2)\cdots(s-p_n)} = \frac{r_1}{s-p_1} + \frac{r_2}{s-p_2} + \cdots + \frac{r_n}{s-p_n}$$

- Multiplying by $(s - p_1)$:

$$\frac{b_0 + b_1 s + \cdots + b_m s^m}{(s-p_2)\cdots(s-p_n)} = r_1 + \frac{r_2(s-p_1)}{s-p_2} + \cdots + \frac{r_n(s-p_1)}{s-p_n}$$

- For $s = p_1$:

$$r_1 = \left[\frac{b_0 + b_1 s + \cdots + b_m s^m}{(s-p_2)\cdots(s-p_n)}\right]_{s=p_1}$$

- In the same way:

$$r_i = \left[\frac{b_0 + b_1 s + \cdots + b_m s^m}{(s-p_1)\cdots(s-p_{i-1})(s-p_{i+1})\cdots(s-p_n)}\right]_{s=p_i} \quad i = 1, \ldots, n$$

Inverse Laplace Transform (4)

Example: Find the inverse Laplace transform of:

$$G(s) = \frac{s^2 - 2}{s^3 + 3s^2 + 2s}$$

- $G(s)$ can be written as:

$$G(s) = \frac{s^2 - 2}{s(s+1)(s+2)} = \frac{r_1}{s} + \frac{r_2}{s+1} + \frac{r_3}{s+2}$$

- r_1, r_2 and r_3 can be calculated from:

$$r_1 = \left[\frac{s^2 - 2}{(s+1)(s+2)}\right]_{s=0} = -1$$

$$r_2 = \left[\frac{s^2 - 2}{s(s+2)}\right]_{s=-1} = 1$$

$$r_3 = \left[\frac{s^2 - 2}{s(s+1)}\right]_{s=-2} = 1$$

Inverse Laplace Transform (5)

- Therefore:

$$G(s) = -\frac{1}{s} + \frac{1}{s+1} + \frac{1}{s+2}$$

- Consequently:

$$\begin{aligned} g(t) &= -u(t) + e^{-t}u(t) + e^{-2t}u(t) \\ &= [-1 + e^{-t} + e^{-2t}]u(t) \end{aligned}$$

- The ROC is given by:

$$\begin{aligned} ROC &= ROC1 \cap ROC2 \cap ROC3 \\ &= \{s \mid \mathrm{Re}(s) > 0\} \cap \{s \mid \mathrm{Re}(s) > -1\} \cap \{s \mid \mathrm{Re}(s) > -2\} \\ &= \{s \mid \mathrm{Re}(s) > 0\} \end{aligned}$$

Inverse Laplace Transform (6)

Case 2: Assume that:

- $m < n$.
- One pole is repeated (for example, the first pole has multiplicity 2).

$$G(s) = \frac{b_0 + b_1 s + \cdots + b_m s^m}{(s - p_1)^2 (s - p_3) \cdots (s - p_n)}$$

In this case, the partial fraction expansion of $G(s)$ can be written as:

$$G(s) = \frac{r_1}{(s - p_1)^2} + \frac{r_2}{s - p_1} + \frac{r_3}{s - p_3} + \cdots + \frac{r_n}{s - p_n}$$

$r_3, \ldots, r_n$ can be calculated in the same way as before:

$$r_i = \left[\frac{b_0 + b_1 s + \cdots + b_m s^m}{(s - p_1)^2 (s - p_3) \cdots (s - p_{i-1})(s - p_{i+1}) \cdots (s - p_n)} \right]_{s = p_i}$$

Inverse Laplace Transform (7)

Multiplying by $(s-p_1)^2$:

$$\frac{b_0+b_1s+\cdots+b_ms^m}{(s-p_3)\cdots(s-p_n)}=r_1+r_2(s-p_1)+\frac{r_3(s-p_1)^2}{s-p_3}+\cdots+\frac{r_n(s-p_1)^2}{s-p_n}$$

For $s=p_1$:

$$r_1=\left[\frac{b_0+b_1s+\cdots+b_ms^m}{(s-p_3)\cdots(s-p_n)}\right]_{s=p_1}$$

Finally:

- We choose a specific value of s.
- We solve for r_2.

Inverse Laplace Transform (8)

Example: Find the inverse Laplace transform of:

$$G(s) = \frac{1}{s^2(s+1)}$$

- $G(s)$ can be written as:

$$G(s) = \frac{r_1}{s^2} + \frac{r_2}{s} + \frac{r_3}{s+1}$$

- r_1 and r_3 can be calculated from:

$$r_1 = \left[\frac{1}{s+1}\right]_{s=0} = 1$$

$$r_3 = \left[\frac{1}{s^2}\right]_{s=-1} = 1$$

Inverse Laplace Transform (9)

- Choose any value of $s \neq 0, -1$. For example, $s = 1$:

$$\frac{1}{s^2(s+1)} = \frac{1}{s^2} + \frac{r_2}{s} + \frac{1}{s+1}$$

$$\text{For } s = 1: \quad \frac{1}{2} = 1 + r_2 + \frac{1}{2}$$

$$\Rightarrow \quad r_2 = -1$$

- Therefore:

$$G(s) = \frac{1}{s^2} - \frac{1}{s} + \frac{1}{s+1}$$

- Finally, given that $\frac{t^{n-1}}{(n-1)!}u(t) \leftrightarrow \frac{1}{s^n}$, then $tu(t) \leftrightarrow \frac{1}{s^2}$ and:

$$g(t) = tu(t) - u(t) + e^{-t}u(t)$$

$$= [t - 1 + e^{-t}]u(t)$$

Inverse Laplace Transform (10)

When a pole (example p_1) has a multiplicity k, $G(s)$ takes the form:

$$G(s) = \frac{b_0 + b_1 s + \cdots + b_m s^m}{(s - p_1)^k (s - p_{k+1}) \cdots (s - p_n)}$$

In this case, the partial fraction expansion takes the form:

$$G(s) = \frac{r_1}{(s - p_1)^k} + \frac{r_2}{(s - p_1)^{k-1}} + \cdots + \frac{r_k}{s - p_1} + \frac{r_{k+1}}{s - p_{k+1}} + \cdots + \frac{r_n}{s - p_n}$$

In this case:

- $r_1, r_{k+1}, \ldots, r_n$ can be calculated as before.
- $r_2, \ldots, r_k$ can be calculated by fixing $k - 1$ values of s and solving the resulting $k - 1$ equations (in $k - 1$ unknowns).

Inverse Laplace Transform (11)

Example: Find the inverse Laplace transform of:

$$G(s) = \frac{1}{s(s+1)^3}$$

- $G(s)$ can be written as:

$$G(s) = \frac{r_1}{s} + \frac{r_2}{(s+1)^3} + \frac{r_3}{(s+1)^2} + \frac{r_4}{s+1}$$

- r_1 and r_2 follow from:

$$r_1 = \left[\frac{1}{(s+1)^3}\right]_{s=0} = 1$$

$$r_2 = \left[\frac{1}{s}\right]_{s=-1} = -1$$

Inverse Laplace Transform (12)

- r_3 and r_4 can be determined by choosing any two values of s (different from 0 and -1). For example, $s = 1, 2$

$$\frac{1}{s(s+1)^3} = \frac{1}{s} - \frac{1}{(s+1)^3} + \frac{r_3}{(s+1)^2} + \frac{r_4}{s+1}$$

$$\text{For } s = 1: \quad \frac{1}{8} = 1 - \frac{1}{8} + \frac{r_3}{4} + \frac{r_4}{2}$$

$$\text{For } s = 2: \quad \frac{1}{54} = \frac{1}{2} - \frac{1}{27} + \frac{r_3}{9} + \frac{r_4}{3}$$

- Solving for r_3 and r_4:

$$r_3 = -1 \ ; \ r_4 = -1$$

Inverse Laplace Transform (13)

- Therefore:

$$G(s) = \frac{1}{s} - \frac{1}{(s+1)^3} - \frac{1}{(s+1)^2} - \frac{1}{s+1}$$

- The relation $\frac{t^{n-1}}{(n-1)!}e^{-\alpha t}u(t) \leftrightarrow \frac{1}{(s+\alpha)^n}$ implies that:

$$g(t) = \left[1 - \frac{t^2}{2}e^{-t} - te^{-t} - e^{-t}\right]u(t)$$

Inverse Laplace Transform (14)

Case 3: Assume that:

- $m \geq n$ ($G(s)$ is called nonproper).

$G(s)$ must first be written as:

$$G(s) = \frac{b_0 + b_1 s + \cdots + b_m s^m}{a_0 + a_1 s + \cdots + a_n s^n} = \frac{b(s)}{a(s)}$$
$$= c(s) + \frac{d(s)}{a(s)}$$

where:

$$c(s) = c_0 + c_1 s + \cdots + c_{m-n} s^{m-n}$$
$$d(s) = d_0 + d_1 s + \cdots + d_k s^k \quad k < n$$

In this case:

- A partial fraction expansion can be applied on $d(s)/a(s)$ (since $k < n$).

Inverse Laplace Transform (14)

Example: Find the inverse Laplace transform of:

$$G(s) = \frac{5s+3}{s+1}$$

- $G(s)$ can be written as:

$$G(s) = \frac{5s+5-2}{s+1} = \frac{5(s+1)-2}{s+1} = 5 - \frac{2}{s+1}$$

- Consequently:

$$g(t) = 5\delta(t) - 2e^{-t}u(t)$$

Inverse Laplace Transform (15)

Example: Find the inverse Laplace transform of:

$$G(s) = \frac{5s^2}{s^2 + 13s + 36}$$

- $G(s)$ can be written as:

$$G(s) = \frac{5(s^2 + 13s + 36) - 5(13s + 36)}{s^2 + 13s + 36}$$
$$= 5 - 5\frac{13s + 36}{s^2 + 13s + 36} \triangleq 5 - 5G_1(s)$$

- $G_1(s)$ can be written as:

$$G_1(s) = \frac{13s + 36}{s^2 + 13s + 36} = \frac{13s + 36}{(s + 4)(s + 9)}$$
$$= \frac{r_1}{s + 4} + \frac{r_2}{s + 9}$$

Inverse Laplace Transform (16)

- r_1 and r_2 are given by:

$$r_1 = \left[\frac{13s+36}{s+9}\right]_{s=-4} = -\frac{16}{5}$$
$$r_2 = \left[\frac{13s+36}{s+4}\right]_{s=-9} = \frac{81}{5}$$

- Therefore:

$$G(s) = 5\left[1 + \frac{16}{5}\frac{1}{s+4} - \frac{81}{5}\frac{1}{s+9}\right]$$
$$= 5 + \frac{16}{s+4} - \frac{81}{s+9}$$

- Consequently:

$$g(t) = 5\delta(t) + \left[16e^{-4t} - 81e^{-9t}\right]u(t)$$

Inverse Laplace Transform (17)

Example: Find the inverse Laplace transform of:

$$G(s) = \frac{s^2}{(s^2+1)(s+1)}$$

- We observe that $G(s)$ has complex poles ($\pm j$). Consequently, we write $G(s)$ under the following form:

$$G(s) = \frac{s^2}{(s^2+1)(s+1)} = \frac{r_1 s + r_2}{s^2+1} + \frac{r_3}{s+1}$$

where, as before:

$$r_3 = \left[\frac{s^2}{s^2+1}\right]_{s=-1} = \frac{1}{2}$$

Inverse Laplace Transform (18)

- r_1 and r_2 can be determined by choosing any two values of s (different from $-1, \pm j$) and solving the system of 2 equations in 2 unknowns:

$$s = 0 \;\Rightarrow\; 0 = r_2 + r_3 \;\Rightarrow\; r_2 = -r_3 = -\frac{1}{2}$$
$$s = 1 \;\Rightarrow\; \frac{1}{4} = \frac{r_1}{2} + \frac{r_2}{2} + \frac{r_3}{2} \;\Rightarrow\; r_1 = \frac{1}{2}$$

- Consequently:

$$G(s) = \frac{1}{2}\frac{s}{s^2+1} - \frac{1}{2}\frac{1}{s^2+1} + \frac{1}{2}\frac{1}{s+1}$$
$$\Rightarrow\; g(t) = \frac{1}{2}\left[\cos(t) - \sin(t) + e^{-t}\right] u(t)$$

Inverse Laplace Transform (19)

Example: Find the inverse Laplace transform of:

$$G(s) = \frac{1}{s^2 + 5s + 7}$$

- We observe that $G(s)$ has complex poles. Consequently, we write $G(s)$ under the following form:

$$G(s) = \frac{1}{s^2 + 5s + \frac{25}{4} - \frac{25}{4} + 7} = \frac{1}{\left(s + \frac{5}{2}\right)^2 + \frac{3}{4}}$$

$$= \frac{2}{\sqrt{3}} \frac{\frac{\sqrt{3}}{2}}{\left(s + \frac{5}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

- Consequently:

$$g(t) = \frac{2}{\sqrt{3}} e^{-\frac{5}{2}t} \sin\left(\frac{\sqrt{3}}{2} t\right) u(t)$$

Solving differential equations (1)

- Laplace transform is a very powerful tool for solving differential equations with initial conditions.
- Since the differentiation in time is equivalent to the multiplication by s in the complex-frequency domain:
  - Solving a differential equation is transformed into solving an algebraic equation.

Solving differential equations (2)

Example: Solve the differential equation:

$$\frac{\mathrm{d}^2 x(t)}{\mathrm{d}t^2} + 7\frac{\mathrm{d}x(t)}{\mathrm{d}t} + 12x(t) = 0$$

for $t > 0$, subject to the initial conditions:

$$x(0^-) = 2 \;\; ; \;\; \left[\frac{\mathrm{d}x(t)}{\mathrm{d}t}\right]_{t=0^-} = -4$$

- Following from the time differentiation property of the Laplace transform:

$$x'(t) = \frac{\mathrm{d}x(t)}{\mathrm{d}t} \leftrightarrow sX(s) - x(0^-)$$

$$x''(t) = \frac{\mathrm{d}^2 x(t)}{\mathrm{d}t^2} \leftrightarrow s^2 X(s) - sx(0^-) - x'(0^-)$$

Solving differential equations (3)

- Consequently:

$$
\begin{aligned}
& s^2X(s) - sx(0^-) - x'(0^-) + 7\left[sX(s) - x(0^-)\right] + 12X(s) = 0 \\
\Rightarrow\ & s^2X(s) - 2s + 4 + 7sX(s) - 14 + 12X(s) = 0 \\
\Rightarrow\ & X(s) = \frac{2s+10}{s^2+7s+12}
\end{aligned}
$$

- $X(s)$ can be written as:

$$X(s) = \frac{2s+10}{(s+3)(s+4)} = \frac{r_1}{s+3} + \frac{r_2}{s+4}$$

- r_1 and r_2 can be calculated from:

$$
\begin{aligned}
r_1 &= \left[\frac{2s+10}{s+4}\right]_{s=-3} = 4 \\
r_2 &= \left[\frac{2s+10}{s+3}\right]_{s=-4} = -2
\end{aligned}
$$

Solving differential equations (4)

- Therefore:

$$X(s) = \frac{4}{s+3} - \frac{2}{s+4}$$

- Consequently:

$$x(t) = \left[4e^{-3t} - 2e^{-4t}\right] u(t)$$

- Checking the result:
  - $x(0^-) = 2$.
  - $x'(0^-) = -12 + 8 = -4$.

System Stability

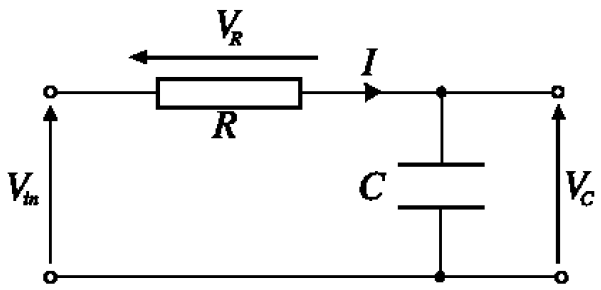
A LTI system is stable if all the poles lie in the open left-half s-plane. The ω axis is not included in the open left-half-plane (LHP).

Proof:

- When passing from the s-domain to the time domain (by applying the inverse Laplace transform), the impulse response of a LTI system will contain terms having the form $e^{p_i t}$ where p_i is the i-th pole.
- For the response to a small perturbation to decay to zero as $t \to +\infty$, $\mathrm{Re}(p_i)$ must be negative for all values of i.
- For example:
  - $\frac{1}{s^2+s+1}$ is stable since the poles $p_1 = e^{j\frac{2\pi}{3}}$ and $p_2 = e^{-j\frac{2\pi}{3}}$ lie in the open LHP.
  - $\frac{1}{s^2+1}$ is not stable since the poles $p_1 = j$ and $p_2 = -j$ lie on the ω axis (not in the open LHP).
  - $\frac{1}{s^2+3s+2} = \frac{1}{(s+1)(s+2)}$ is stable.

First Order Filters (1)

- A series RC circuit can serve as a low pass filter.



- In the time domain:

$$v_{in}(t) = v_R(t) + v_C(t) = Ri(t) + v_C(t) = RC\frac{\mathrm{d}v_C(t)}{\mathrm{d}t} + v_C(t)$$

- In the s-domain:

$$V_{in}(s) = RC\left[sV_C(s) - v_C(0^-)\right] + V_C(s)$$

- Assuming that the capacitor was initially uncharged ($v_C(0^-) = 0$):

$$H(s) = \frac{V_C(s)}{V_{in}(s)} = \frac{1}{RCs+1} = \frac{1/\tau}{s+1/\tau} \quad ; \quad \tau = RC$$

First Order Filters (2)

$$H(s) = \frac{1/\tau}{s + 1/\tau} \quad ; \quad \tau > 0$$

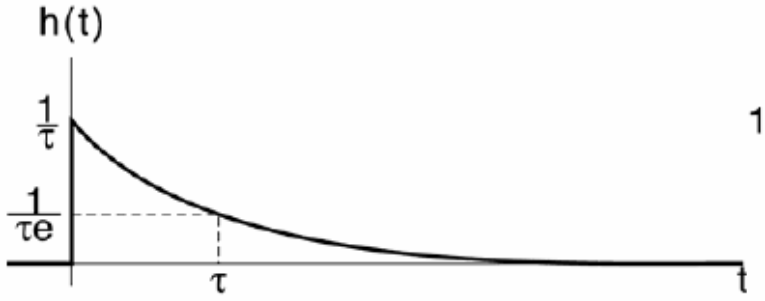
- The impulse response of the system is given by:

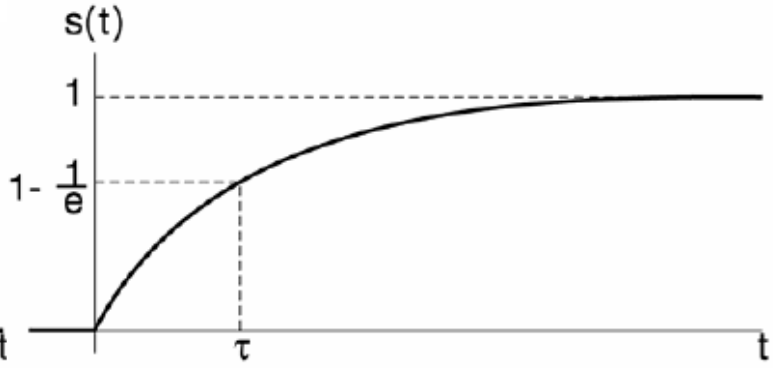
$$h(t) = \frac{1}{\tau} e^{-\frac{t}{\tau}} u(t)$$

- The response to a unit step can be calculated from:

$$S(s) = \frac{1}{s} H(s) = \frac{1/\tau}{s(s + 1/\tau)} = \frac{r_1}{s} + \frac{r_2}{s + 1/\tau} = \frac{1}{s} - \frac{1}{s + 1/\tau}$$

$$\Rightarrow s(t) = [1 - e^{-\frac{t}{\tau}}] u(t)$$





First Order Filters (3)

$$H(j\omega) = \frac{1/\tau}{j\omega + 1/\tau}$$

- The magnitude of $H(j\omega)$ verifies the relation:

$$|H(j\omega)| = \frac{1/\tau}{\sqrt{\omega^2 + 1/\tau^2}} = \begin{cases} 1, & \omega = 0; \\ 1/\sqrt{2}, & \omega = 1/\tau; \\ 1/\omega\tau, & \omega \gg 1/\tau. \end{cases}$$

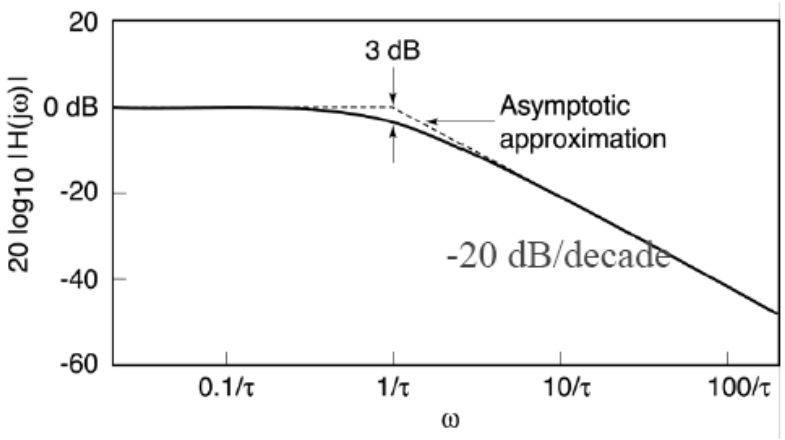
- It is more convenient to express $|H(j\omega)|$ in decibels (dB):

$$|H(j\omega)|_{\text{dB}} = 20 \log_{10} |H(j\omega)| \ \ \text{dB}$$

  - In this case, each time $|H(j\omega)|$ is multiplied by 10, $|H(j\omega)|_{\text{dB}}$ is increased by $20 \log_{10} 10 = 20$ dB.
- For $\omega = 1/\tau$, $|H(j\omega)|_{\text{dB}} = 20 \log_{10}(1/\sqrt{2}) \approx -3$ dB.

First Order Filters (4)

- For $\omega \gg 1/\tau$:

$$
\begin{aligned}
|H(j\omega)|_{\text{dB}} &\approx 20\log_{10}(1/\omega\tau) \\
|H(j10\omega)|_{\text{dB}} &\approx 20\log_{10}(1/10\omega\tau) \\
&= 20\log_{10}(1/\omega\tau) + 20\log_{10}(1/10) \\
&= |H(j\omega)|_{\text{dB}} - 20
\end{aligned}
$$

  - $|H(j\omega)|_{\text{dB}}$ decreases by 20 dB when ω is multiplied by 10.
  - The slope of $|H(j\omega)|_{\text{dB}}$ for $\omega \gg 1/\tau$ is said to be equal to -20 dB/decade.
- A Bode diagram corresponds to plotting $|H(j\omega)|_{\text{dB}}$ against the logarithm of frequency.
- On the x-axis, 1 unit is equivalent to a multiplication by 10.